UNIVERSITÉ CÔTE D'AZUR

ÉCOLE DOCTORALE
SCIENCES ET TECHNOLOGIES DE L'INFORMATION ET DE LA COMMUNICATION

THÈSE DE DOCTORAT

Tritre francais

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*À toi lecteur <3 :**

Tritre francais

Mots-clés : Pâtisserie, Macaron, Ladurée.

English Title

Abstract

Macarons are so good.

Keywords: Pastry, Macaron, Ladurée.

Acknowledgement

Merci !

Contents

Annexes

Chapter 1

Introduction

1.1 Prologue

Automation everywhere in CS, we want to set up boilerplates to help our development.

Examples:

- Continuous integration
- Macros are everywhere
- Artifitial intelligence

We delegate tasks to some tools.

expert system $\rightarrow$

example plus frappant:

artificial intelligence: based on probability but not reproductible, not predicatable

several other examples in computer sciences:

...

1.2 Automation is everywhere

In the domain of formal proof automation is everywhere, and it is formally called elaboration **??**

In formal proof, proofs are verbose and have lot of details: types, coercions, canonical structures, type-classes.

In theorem provers the user benefit of notations, implicit inference and tactics to ease his job.

1.2.1 Probabilistic reasoning: artificial intelligence

1.2.2 Automated reasoning: elaboration

Simply typed lambda calculus is an exmaple of automation: an algo checks if ...

1.3 Contributions

In this manuscript we focus on an alternative type-class solver written in the elpi meta-langauge.

We also talk about a formalized tool. . .

1.3.1 Type-class inference

1.3.2 Static determinacy checking

CHAPTER 2

State of the Art

2.1 The Rocq proof assistant

2.1.1 The basis of the langauge

2.1.2 Elaboration

2.1.3 Ad-hoc polymorphism: Type Classes

2.2 The Elpi language

2.2.1 Controlling backtracking: the ! operator

2.2.2 Hints about unification

2.2.3 Rocq-Elpi: Rocq and Elpi interleaving

2.2.4 CHR: goal suspension

Tactics, coq-elpi

CHAPTER 3

Type classes resolution in Elpi

3.1 Why typeclass resolution in Elpi?

[1]

Rule base automation!

3.2 From Rocq goals to Elpi proofs

3.2.1 Rule encoding

The hooks

3.2.2 Goal encoding

3.3 Alternative search indexing

3.3.1 Discrimination trees

Discrimination trees

TODO: do some benchmarks

Discrimination trees: opt for lists and tree depths.

3.3.2 Tabling

3.4 Helping hihger-order unification

3.4.1 η-equivalent terms

3.4.2 β-equivalent terms

[2]

CHAPTER 4

Determinacy checking for Elpi

4.1 Why determinacy?

4.2 Implementation

4.2.1 Rule checking

4.2.2 Mutual exclusion

Using discrimination tree

4.2.3 Caveats

For example, mutual exclusion is not perfect: over aproximation of unification

Rule checking ignores flexible rule loading

4.3 Formalization in Rocq

4.3.1 The langauge

Input must come before outputs

Inputs must come before outputs, with Disc tree this is not mandatory, since unification is weaker.

EG:

```
pred g o:int, o:int.

f 3 3 (g X X).
f 3 4 (g 3 4).

main:-
  f X Y (g X Y).
```

In the code above, both rules can be used with the query, but, the discrimination tree, save the world

4.3.2 The tree semantics

4.3.3 The stack semantics

4.3.4 Semantics equivalence

4.3.5 The main theorem

Chapter 5

Conclusion and Perspectives

[3]

$$
\begin{tikzcd}
t \arrow[r, "\text{xx } t"] \arrow[d, "\text{t2l}"] & t' \arrow[d, "\text{t2l}"] \\
a \arrow[r, "\text{yy } a"'] \arrow[ur] & a'
\end{tikzcd}
$$

```
func p int -> int.
```

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[3] Davide Fissore and Enrico Tassi. Determinacy checking for elpi: an higher-order logic programming language with cut. In Nada Amin and Joaquín Arias, editors, *Practical Aspects of Declarative Languages*, pages 77–95, Cham, 2026. Springer Nature Switzerland.

List of Figures

Definition’s list

Theorem’s list

Appendix

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BLIBLI

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